

NORMAN BUI

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EDUCATION

Carleton University | *Bachelor of Computer Science (Honours), 3.9/4.0 GPA* Graduated Spring 2026
• Coursework: Statistical Modeling, Data Structures & Algorithms, Database Management Systems, Linear Algebra

EXPERIENCE

Electronic Arts (EA) | *Software Developer Intern* May 2025 - Aug 2025
• Improved CI/CD load-test reliability from **35% to 100%** by fixing auth failures and unstable workflows
• Evaluated LLM-powered agentic QA system for reliability, failure modes, and model behavior across load/soak workflows
• Documented failure taxonomy and evaluation findings to drive model-behavior improvements in automated playtesting pipeline

Autodesk | *Software Developer Intern* Jan 2025 - Apr 2025
• Built TypeScript/Java validation and repair infrastructure for Fusion manufacturing data, including an **O(n)** repair algorithm that improved throughput **8x**
• Designed deterministic validation logic for manufacturing data repair workflows
• Deployed GraphQL validation and repair endpoints with **100%** Mocha, Chai, and Sinon coverage

Public Health Agency of Canada | *Data Scientist Intern* Sep 2024 - Dec 2024
• Built LLM-assisted literature review and MDM workflows for global health data with Imperial College London and WHO
• Evaluated Ollama embeddings for RAG retrieval quality and baseline system-tuning metrics
• Prototyped GPT-4o paper collection for scalable PDF retrieval and metadata extraction

Bedarra Corporation | *Student Researcher* Aug 2024 - Oct 2024
• Researched and developed agentic RAG systems to answer queries over obscure programming language corpora (Factor, K)
• Benchmarked embedding models and chunking strategies for retrieval quality over low-resource language corpora
• Implemented LangChain tool-use and retrieval patterns for context-aware, grounded query responses

Health Canada | *Data Scientist Intern* Sep 2023 - Dec 2023
• Developed toxicity classifiers with pandas, NumPy, SciPy, and scikit-learn
• Built Neo4j knowledge-graph pipeline extracting chemicals, diseases, and literature relationships
• Architected TDD scraping framework with unittest, expanding data sources by **113%** with **100%** coverage

PROJECTS

Orbital Refueling Simulator | *Python, Streamlit, scikit-learn, Qwen LoRA, pytest* github.com/nbui23/orbital-refueling
• Modeled **410-second** refueling mission across **9** phases, **9** anomaly classes, and deterministic generation
• Separated hard-limit engineering rules from **9** phase-aware IsolationForest drift models
• Benchmarked deterministic rule alerts against **0-1** advisory ML drift scores across anomaly scenarios
• Fine-tuned Qwen LoRA explainer on synthetic telemetry instructions, improving held-out rubric score to **4.86/5**

TwinQuery | *Python, LangGraph, FastAPI, Streamlit, PostgreSQL/PostGIS, Ollama, RAG, Docker* github.com/nbui23/twinquery
• Built open-source digital-twin query assistant enabling non-technical users to ask plain-English questions over building-stock and retrofit datasets
• Implemented safe Text-to-SQL pipeline with deterministic guardrails, fallback query templates, and PostgreSQL/PostGIS execution over Ottawa building-footprint geometry
• Integrated local RAG over retrofit guidance with hybrid synthesis, grounding LLM answers in both database rows and retrieved document sources

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL
Frameworks & Runtimes: FastAPI, Flask, Streamlit, LangGraph, LangChain, Next.js, Node.js, Spring Boot
AI / ML / Data: LLMs, RAG, LangGraph, LangChain, Ollama, Qwen LoRA, scikit-learn, pandas, NumPy, PyTorch, FAISS
Databases & APIs: PostgreSQL/PostGIS, pgvector, MySQL, SQLite, MongoDB, Firebase, Neo4j, GraphQL
Tools & Practices: Git, Docker, Docker Compose, AWS, CI/CD, Ubuntu, TDD, pytest, unittest, Postman, Agile, SDLC
Spoken Languages: English, French, Vietnamese